

Assignment -02

Python Programming (CAN303)

1. Explain the basic features of Python. Also discuss Python variables, expressions, and the following data types with suitable examples: integer, float, string, Boolean, list, tuple, set, and dictionary.
2. Write a Python program that accepts a student's marks in three subjects, calculates the total and percentage, and displays the grade using conditional statements. Handle appropriate grade conditions using if-elif-else.
3. Write Python programs to demonstrate the use of for and while loops. Use one example to generate a multiplication table and another example to calculate the sum of the first n natural numbers.
4. Create a Python function to determine whether a given number is prime. Then modify the function so that the range of numbers can be processed and all prime numbers in that range are displayed. Clearly demonstrate function parameters and return values.
5. Explain Python strings and any five commonly used string functions with examples. Write a program that accepts a string and displays the number of vowels, consonants, digits, and spaces present in it.
6. Explain the difference between lists, tuples, sets, and dictionaries. Write a Python program that creates one example of each and demonstrates indexing/slicing where applicable, insertion or update, and removal of elements.
7. Explain the concepts of lambda functions, map(), and filter(). Write a Python program using these concepts to create a list of squares of even numbers from a given list of integers.
8. What are list comprehensions, dictionary comprehensions, and set comprehensions? Write suitable Python examples for each. Also state one advantage of comprehensions over writing the equivalent operation using a loop.
9. Using NumPy, create one-dimensional and two-dimensional ndarray objects. Demonstrate array data types, indexing, slicing, transposing, and basic arithmetic operations between an array and a scalar. Show the output.
10. Using NumPy, write a program that creates a numerical array and demonstrates conditional logic, mathematical/statistical methods, basic linear algebra, and random number generation. Include at least one example each of a statistical calculation, a matrix operation, and generation of random numbers.
11. Create the following NumPy array:

```
A = np.array([[10, 20, 30, 40],  
              [50, 60, 70, 80],  
              [90, 100, 110, 120]])
```

Write NumPy statements to:

1. Extract the second row.
2. Extract the first two columns.
3. Extract the bottom-right 2×2 sub-array.
4. Extract every alternate column.
5. Reverse the order of the rows.